Evaluating Classification (預測分類) Results

Confusion matrix (a.k.a error matrix)

		True condition				
	Total population	Condition positive	Condition negative	Prevalence $= \frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) $= \frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	**True positive**, Power	**False positive**, Type I error	Positive predictive value (PPV), Precision $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR−}}$	F_1 score $= \frac{1}{\frac{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}{2}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) $= \frac{\text{FNR}}{\text{TNR}}$		

五大名詞

Accuracy: 在資料不平衡時沒有參考價值，例如疾病患病率 1% ，模型全猜患病就有 99% acc

Precision: 你抓出來說是「犯人」的人裡面，有多少真的是犯人？ Predict positive based

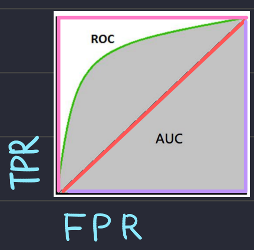
NPV: 在「報告說你沒病」中，真正沒病的人佔多少比例？ Predict negative based

Recall (Sensitivity): 所有的「犯人」裡面，你到底抓到了幾個？Condition positive based

Specificity: 健康的人裡面，排除多少？ Condition negative based

F1 Score: Precision & Recall 的均衡值

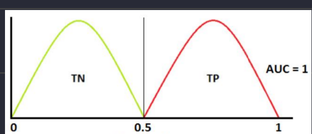
Performance measurement



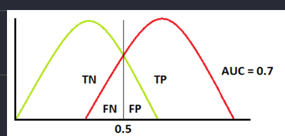
TPR (True Positive Rate) $\frac{Sensitivity}{Recall\ rate}$ → 越高越好

FPR (False Positice Rate) $1 - Specificity$ → 越低越好

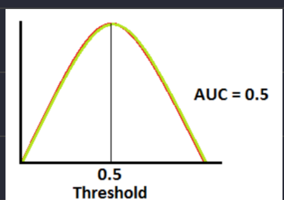
粉色 ROC: AOC=100% 正確



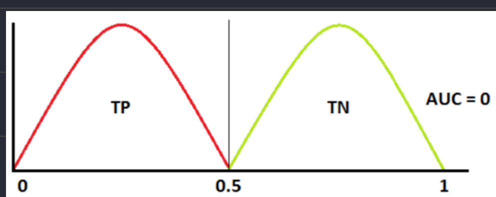
綠色ROC曲線: AOC=70%



紅色 ROC: AOC=50% (用猜的)



紫色 ROC: AOC=0% (結果取反就好)



Confusion Matrix for a 3-Class Classification

Pridicted Values

Actual Values

	Setosa	Versicolor	Virginica
Setosa	16 (cell 1)	0 (cell 2)	0 (cell 3)
Versicolor	0 (cell 4)	17 (cell 5)	1 (cell 6)
Virginica	0 (cell 7)	0 (cell 8)	11 (cell 9)

Tips

True Positive

↳ 表示預測之結果

→ 表示預測結正確性 實際結果就是兩個作 XNOR

直接由得到的實際結果與預測果選 Row 與 Column

For the Setosa class:

True Positive

↳ Positive: 選擇預測正確的 Column → Setosa

→ True: 所以實際情況就是正確的，選正確的 Row → Setosa

Ans: cell 1

True Negative

↳ Negative: 選擇預測錯誤的 Column → Versicolor, Virginia

→ True: 所以實際情況就是錯誤的，選錯誤的 Row → Versicolor, Virginia

Ans: cell 5 + cell 6 + cell 8 + cell 9

False Positive

↳ Positive: 選擇預測正確的 Column → Setosa

→ False: 所以實際情況相反，是錯誤的，選錯誤的 Row → Versicolor, Virginia

Ans: cell 4 + cell 7

False Negative

↳ Negative: 選擇預測錯誤的 Column → Versicolor, Virginia

→ False: 所以實際情況相反，是正確的，選正確的 Row → Setosa

Ans: cell 2 + cell 3

Evaluating Regression（預測數值）Results

Mean Squared Error (MSE) $MSE = \frac{1}{N}\sum_{j=1}^{N}(y_j - \check{y}_j)^2$ → 極度放大較大的誤差

Mean Absolute Error (MAE) $MAE = \frac{1}{N}\sum_{j=1}^{N}|y_j - \check{y}_j|$ → 對離群值（Outliers）不敏感

Root Mean Squared Error (RMSE) $RMSE = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(y_j - \check{y}_j)^2}$ → 稍微收斂 MSE 的放大

R^2 (R-Squared) $SE(Y) = \sum_{i=1}^{N}(y_i - \bar{y})^2$ $SE(line) = \sum_{j=1}^{N}(y_j - \check{y}_j)^2$ $coeff(R^2) = 1 - \frac{SE(line)}{SE(\bar{Y})}$ →模型比「瞎猜平均」強多少

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Tips

True Positive

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直接由得到的實際結果與預測果選 Row 與 Column

For the Versicolor class:

True Positive

↳ Positive: 選擇預測正確的 Column → Setosa

→ True: 所以實際情況就是正確的，選正確的 Row → Setosa

Ans: cell 1

True Negative

↳ Negative: 選擇預測錯誤的 Column → Versicolor, Virginia

→ True: 所以實際情況就是錯誤的，選錯誤的 Row → Versicolor, Virginia

Ans: cell 5 + cell 6 + cell 8 + cell 9

False Positive

↳ Positive: 選擇預測正確的 Column → Setosa

→ False: 所以實際情況相反，是錯誤的，選錯誤的 Row → Versicolor, Virginia

Ans: cell 4 + cell 7

False Negative

↳ Negative: 選擇預測錯誤的 Column → Versicolor, Virginia

→ False: 所以實際情況相反，是正確的，選正確的 Row → Setosa

Ans: cell 2 + cell 3

Data mining Techniques

Types of Data Mining / Machine Learning

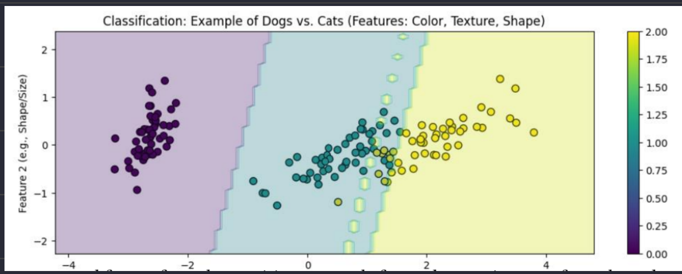
1. Supervised Machine Learning

Pros: 品質高資料有高精確度，並且與訓練模型比較快，也有可解釋性

Cons: 對於沒看過的資料有困難，標記資料費時，沒有辦法隨新資料變通

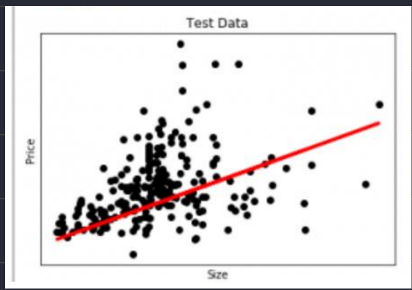
(a) Classification：分類、判斷是非

- Logistic Regression
- Support Vector Machine
- Random Forest
- Decision Tree
- K-Nearest Neighbors (KNN)



(b) Regression：預測、推斷數值

- Random Forest
- Decision tree
- Polynomial Regression
- Linear Regression



2. Unsupervised Machine Learning

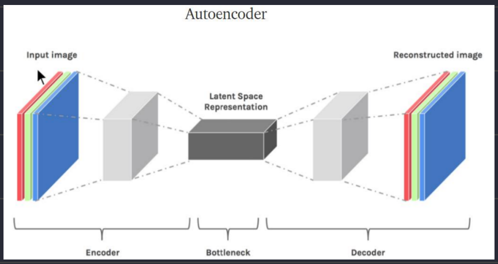
Pros: 尋找隱藏規律與關係、用於異常檢測、資料探索，減少標記的工作量

Cons: 無法衡量模型品質、沒有可解釋性(Cluster Interpretability)

Abnormaly detection：排除異常值

Auto Encoder (自編碼器)

從高維度數據中提取關鍵特徵，或如何降低數據的維度以節省計算資源



(a) Clustering：通過資料特徵分群 e.g. 牙膏與牙刷在浴室用品區

- K-Means Clustering algorithm
- Mean-shift algorithm
- Independent Component Analysis

(b) Association：找出A發生時B也會發生 (A implies B) 的機率 e.g. 買尿布後會買啤酒

- Apriori Algorithm
- FP-growth Algorithm

Association rule learning：使用獨立性強的欄位

例如：帥、鼻子挺、眼睛大 都在描述同一件事

(c) Dimensionality Reduction ：在保持準確性下去除多餘維度、加速效率

- PCA (Principal components analysis)

3. Semi-supervised learning (半監督式學習)

Pros: 比監督式學習更普遍化，可以套用到更廣泛的資料

Cons: 仍需要部分標記資料，且實作上更困難

GANs (生成對抗網路)

a) 生成網路 (generator network) 生成資料

b) 鑑別網路 (discriminator network) 辨識資料

→ 兩個神經網路相互博弈

4. Reinforcement Learning

TD;DR

每一次與環境互動進行學習，並根據反饋的好壞，由機器自行逐步修正、最終得到正確的結果

(a) Q-learning ├─ model-free RL algorithm

The Q-function (是一個紀錄過去經驗的表格) estimates the expected reward of taking a particular action in a given state

(b) SARSA (State-Action-Reward-State-Action) ├─ model-free RL algorithm

Unlike Q-learning, SARSA updates the Q-function for the action that was actually taken, rather than the optimal action.

(c) Deep Q-learning

當情況很多時，不可能用表紀錄，直接擬合一條曲線取代紀錄查表

Positive reinforcement (做事獲得獎勵)

1. Rewards the agent for taking a desired action.
2. Encourages the agent to repeat the behavior.

Examples: Giving a treat to a dog for sitting, providing a point in a game for a correct answer.

Negative reinforcement (做事避免懲罰)

1. Removes an undesirable stimulus to encourage a desired behavior.
2. Discourages the agent from repeating the behavior.

Examples: Turning off a loud buzzer when a lever is pressed, avoiding a penalty by completing a task